

Exact living band

NOT_COFINAL; 2^k commensurability refuted;
 k_z197 is a new ζ -chain death

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Abstract

r447 left a caveat: k_z137 (float-dead at $n = 7511$) stayed positive on exact atoms through 7516, so “last live k_z136 ” was a float statement, while $k_z197 = 2^{10}$ is exactly dead. This note remaps the band on the r447 exact-atom builder. Every sampled window $k_z \geq 137$ is float-dead. Exact k_z230 (generic prime $a = 1259$) dies at the same $n = 1818$ as float, dps 50 = 70 bit-identical — so 2^k anchors are not specially endangered. k_z197 is the first measured true ζ -window positivity break: an *in-chain* death at $n/N \approx 0.931$, not a dead- χ pole overshoot. The predefined repair $a_k = \text{next odd after } 2^k$ dies at $k = 10$. No cofinal living family is found. No RH claim.

Status. No new SATZ. Propositions 1–4 are a named infra census. Ausgang NOT_COFINAL / COMMENSURABILITY_REFUTED / NEXT_ODD_FAILS / ZETA_CHAIN_DEATH / LAST_LIVE_FLOAT_136. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 Band map

Priority skeleton: k_z137 full ($N = 8300$), then 150, 230, 339, 500; float pre-screen on the rest. Exact atoms reuse the r447 builder (mp log of integer prime powers, no float64 cast).

Proposition 1 (Float mesh ends at k_z136). k_z136 ($a = 631$, $N = 1641$) is float-living. Every sampled $k_z \in \{137, \dots, 145\}$ and the stride $\{150, 155, \dots, 500\}$ is float-dead. The first-flip ratio n/N falls from 0.997 (k_z138) through 0.931 (k_z197) and 0.904 (k_z230) toward ~ 0.5 at large k_z .

Proposition 2 (Generic exact death). k_z230 ($a = 1259$, not a 2-power) on exact atoms, dps 50 and 70: first flip $n = 1818 = n_{\text{float}}$, $\eta = -2.2240756817800884 \cdot 10^{-12}$ bit-identical, $\text{rel} = 3.40 \cdot 10^{-6}$, $n/N = 0.904$. The 2^k commensurability hypothesis is REFUTED: a generic anchor dies exactly, at the same place as float.

k_z197 remains EXACT_DEAD at $n = 3788$ (r447). k_z170 dies at $n = 5515$ (float-concordant, $n/N = 0.956$). k_z137 dies at $n = 8283$ ($n/N = 0.998$), not at the float pivot 7511 — a location-lie, not a resurrection. Last live remains k_z136 under exact atoms. The band is not cofinal.

2 k_z197 anatomy

Four coordinates, versus living k_z136 and dead $\text{CHI}_3\text{-15}$:

	q_N	$ Z_{\text{loc}} $	$q^\dagger$ / sch	break
k_z197	-1.682 (pack), 1.794 (C2)	0.033	$q_{\text{pos}}^\dagger = 0.968$	in-chain $n/N = 0.931$
k_z136	0.220	0.306	$q^\dagger = 0.972$, sch = -0.028	none (living)
CHI ₃ -15	1.277	—	$q^\dagger = 1.023$, sch = +0.023	none: $n_{\text{flip}} = 0$, $q > 1$

Proposition 3 (A new ζ -death type). *k_z197 is not a dead- χ world. Dead χ keep a positive chain and die by pole overshoot ($q^\dagger > 1$ at the edge). k_z197 breaks inside the signed mass chain at $n/N \approx 0.931$; $|Z_{\text{loc}}| = 0.033$ is living-small, while $|Z| = 1.132$ overflows. This is a chain-positivity death on a ζ -window, the first measured.*

The 2^{10} anchor $a = 1024$ is log-2 commensurable with the dyadic fold, but Proposition 2 shows the same in-chain death at a generic prime. The resonance is not the cause.

3 Repair family

Predefined, no sign-peeking: a_k = the next odd integer after 2^k that the sieve marks, k_z via the same m -formula as r397 ($\alpha = \log a$, $\Delta = (u_{k_z+1} - u_{k_z})/(2\nu)$). Then $a \rightarrow \infty$ and $\Delta \rightarrow 0$ as $k \rightarrow \infty$.

Proposition 4 (Next-odd fails). *Float (and, at $k = 10$, the neighbour k_z198): the next-odd family lives for $k \leq 9$ and dies at $k = 10$ ($p = 1031$, first $n = 10121$). Two other predefined families fail earlier: $3 \cdot 2^{k-1}$ at $k = 9$, 3^k at $k = 6$. No living δ -sequence past $k = 9$ is obtained; the r445 slice floor ($\delta_\infty = +0.02741$) stands.*

Lean sketch (not built): the r397 selected family remains a finite living prefix $k \leq 9$. No revision of `rh_of_frequently_selected`. Other cofinal meshes stay a reviewer fork.

4 Float concordance

Exact-concordant deaths: k_z170 , k_z197 , k_z230 (float first = exact first, dps-stable at k_z230). k_z137 : float first 7511, exact first 8283 ($n/N = 0.998$, $\eta \sim -3 \cdot 10^{-15}$) — one location-lie in four exact deaths. Living side ($k = 5, 9$, k_z136) agrees.

Claim boundary

Nothing here is evidence for or against the Riemann hypothesis. The identities are a finite recurrence census on named windows. The frequently-quantifier gap stays open for families not measured here.

References

- [1] Selected sequence; $a_k \rightarrow \infty$, $\Delta_k \rightarrow 0$.
- [2] Deep builder; slice floor $\delta_\infty = +0.02741$.
- [3] Float64-atom REAL; last living k_z136 .
- [4] Exact-atom EXACT_DEAD at $k = 10$.